CHAPTER III

NYĀYA-VAIŚEṢIKA ON ARTHĀPATTI

Since Arthāpatti is primarily concerned with the framing of hypothesis and since hypothesis relate to something as yet unperceived, Arthāpatti needs to be regarded as a source of parokṣa (non-perceptual) and not aparokṣa (perceptual) cognition. So the question of its reducibility to perception cannot arise. And its reducibility to either testimony or comparison, on account of their being sources of parokṣa cognition, is obviously out of the question. The only source of parokṣa cognition, its reducibility to which may be characterized by some measure of plausibility, should be none other than inference. But as we have seen earlier, both Prabhākara and Kumārila have argued the distinctness of Arthāpatti from inference in their respective ways. The Advaita Vedānta also does the same in the following manner. This school of philosophy, be it noted in the beginning, holds that in the case of inference strictly so called, the universal major premise must be based on positive concomitance (anvayavyāpti), and that the inference in which negative concomitance (Vyatirekavyāpti) constitutes the basis of its universal major premise is really no inference as such, but is another name for Arthāpatti. Judged in the light, Arthāpatti, as the Advaita Vedānta argues, is distinct from inference

for the simple reason that a universal major premise based on positive concomitance is not available in its case, the proposition, for example, "whoever there is stoutness (pīnatvaim) there is the condition of eating at night (rātṛibhojanam)" being contrary to fact. But this really brings to light the crux of the whole situation by leaving behind the demand for a fresh enquiry into the possibility of inference with a universal major premise based on negative concomitance (Vyatireka-Vyāpti). Hence, arises the necessity for the consideration of the attempts made by the Vaiśeṣika, the Nyāya and the Sāṁkhya to show that Arthāpatti, in the final analysis, is a form of inference.

Let us begin by observing that Prabhākara's attempt to account for the distinction between Arthāpatti and Anumāna with reference to his admission of the presence of an element of doubt in the former and the absence of it in the latter proves a failure in the light of Kumārila's finding that Arthāpatti, as a matter of fact, does not have to bear the burden of any such thing as doubt. Even so, Kumārila is as insistent on the recognition of this distinction as is Prabhākara, and finds the reason for this recognition in his view that Arthāpatti differs from inference in that, whereas the former involves the conflict between two known facts, the latter is free from

such involvement. But then, any attempt to argue the distinction between the two sources of cognition under consideration, not with reference to their respective peculiarities as ways of cognizing, but reference to the element or elements supposed to be involved in them, is undoubtedly superficial and cannot really serve the purpose which it is intended to serve. This seems to have been realized by the Advaita Vedānta as is evident from the fact, that, instead of undertaking the useless task of ascertaining the factors likely to be involved in Arthāpatti, it straight away takes notice of the peculiarity of this way of cognizing and accordingly states that it is none but the act of framing hypothesis with a view to explaining situations which call for explanation. Thus has the Advaita Vedānta, as it seems to me, offered the most realistic interpretation of the nature of Arthāpatti, which as will be explained later, hardly leaves any scope for asking the question whether this source of cognition is reducible to any other and especially inference.

As previously indicated, the Advaita Vedānta, while dismissing the possibility of the interpretation of Arthāpatti as identical with that kind of inference in which the universal major premise is based upon positive concomitance, does not rule out, but on the contrary, admits the possibility of its being regarded as the same as the kind of inference in whose case the universal major premise

is based on negative concomitance. Even so, it seeks to rescue Arthāpatti from its absorption in inference by declaring the inference of the latter kind to be nothing but Arthāpatti in disguise. But this is too simple and easy way of vindicating the independence of this way of cognising to produce any salutary effect upon the Nyāya-Vaiśeṣika school of philosophy which, consistently with their respective epistemological positions, are intent upon establishing the identity of Arthāpatti with inference. It is, therefore, no wonder that all of them should try to show that Arthāpatti is indistinguishable from inference in one form or another. It is, however, the philosophers belonging to the Vaiśeṣika and the Nyāya schools who have taken the greatest interest in the performance of this task.

A. GAUTAMA AND VĀTSYĀYANA ON ARTHĀPATTI

The Mimāṁsaka's was not the only account of Arthāpatti current in the early schools; and it was not the original meaning of the term. The notion is discussed in that passage of the Nyāya-sūtra[1], which argues against the separate status of supposed pramāṇa's other than four enumerated in Nyāya sūtra I, i.3; and the account there given, as interpreted by Vātsyāyana, is not identical with the account given by Śabara.

The topic of Arthāpatti is introduced by Gautama in the aphorism 2.2.1 by way of an objection that his classification of sources of knowledge is inadequate, because he has left out Arthāpatti (and three sources of knowledge). Gautama defends his classification in the following aphorism by saying (among other things) that Arthāpatti is included in inference (Anumāna). His own statement of objection is: "(The sources of knowledge) are not four, because Aitihya, Arthāpatti, Sambhava, and Abhāva are (additional) sources of knowledge". The reply: "Since Aitihya (tradition) is not different from Śabda (authoritative statement) and since Arthāpatti, Sambhava, and Abhāva are not different from Anumāna (inference), there is no refutation (of the view that the sources of knowledge are four, namely, perception, inference, analogy and authoritative statement)". From this objection and reply we know that already before the time of Gautama there were philosophers who subscribed to a narrow view of inference and excluded Arthāpatti and so on from it as distinct sources of knowledge as against that Gautama is subscribing to a broad conception of inference that would be inclusive of Arthāpatti and so on.

In the aphorism following those mentioned earlier, Gautama has discussed to question of the validity or invalidity of Arthāpatti. But he has not explained what

is meant by Arthāpatti. For such an explanation we have to look at Vātsyayana, whose definition of Arthāpatti is as follows: "Where from a proposition stated another proposition follows as a necessary consequences, that is Arthāpatti."[2] This definition of Arthāpatti makes it applicable to any valid deductive reasoning with a single premise and single conclusion. Vātsyāyana has not emphasized that there should be a single premise and single conclusion. He has merely put Artha in the singular number. Hence one cannot rule out the possibility that there could be more than one premise. However, in both of the examples available from Vātsyāyana, there is only one premise. Thus it is probable that what is meant by Arthāpatti is a deductive reasoning with a single premise.

Vātsyāyana's definition of Arthāpatti does not make it clear on what the implication, between the premise and the conclusion, is based. His other remarks and examples, however, help us to see what the basis is. We will first look at his example. "For instance, what is implied in the statement that when there is no cloud it does not rain? (The implication is) that it rains where there are clouds".[3] i.e. the object cognised through implication is that the production of effect, rain, is limited to the existence of the cause, clouds. If not A, not B: ergo if B, then A. Vātsyāyana further characterises this process

as 'apprehending from opposition what is not stated'.[4] He says shortly afterwards: "from the statement that in the absence of the cause the effect is not produced, we arrive by implication at what is related to thus on its opposite, namely that the effect is produced in the presence of the cause. For from a negative comes the opposed affirmative".[5]

It may be noticed that the premise and the conclusion in this argument are conditional statements. We have seen that Gautama and Vātsyāyana were concerned in disjunctive syllogism with reasoning involving the two connectives of negation and disjunction. The present specimen shows that reasoning involving yet another connective, namely, the conditional, occupied their attention. Moreover, in this case both the premise and the conclusion are unmistakably propositions, and hence the reasoning belongs, without doubt, to the logic of propositions.

Gautama has declared that reasonings of this kind are to be included in inference. Regarding that, Vātsyāyana's comments are as follows: "Inference is knowledge through the given of what is not given, but related (to the given)..... Since Arthāpatti is knowledge of a proposition which is not stated obtained from the precise understanding of the meaning of a sentence by means of the

relation of opposition, it is definitely inference".[6] According to this quotation, Arthāpatti is knowledge from the understanding of the meaning of a sentence. In modern terminology this amounts to saying that Arthāpatti is analytic knowledge and not knowledge obtained by any empirical means.

Moreover, the proposition deduced is said to be related by way of 'opposition' to the premise. Vātsyāyana has clarified what is meant by 'opposition' (pratyanikabhāva) as follows: "If there is no cause, there is no effect from the meaning of this sentence is obtained the proposition related by way of opposition that 'if there is cause, there is effect'. The positive is the opposite (pratyanika) of the negative".[7] What is meant is that 'there is no cause' is related by way of opposition to 'there is cause', and similarly 'there is no effect', to 'there is effect'. In the proposition 'there is no cause' the sanskrit sentence shows that what is negated is the proposition 'there is cause' and similarly in the case of 'there is no effect'. Thus we know from this example that a proposition and its negation are related by way of opposition.

It is now clear that the implication on which the deduction of 'if there are clouds, there is rain' from

'if there are no clouds, there is no rain' is based on the relation of opposition. What is required is that the conclusion should contain negation of the constituent simple propositions in the premise.

We have another specimen of Arthāpatti in Vātsyāyana's comments[6] "after stating 'non-eternal because of being originated' it is obtained from the meaning that..... what is eternal is non-originated..... this ..... by Arthāpatti". Here 'non-eternal because of being orginated' is certainly not intended to be the premise as it stands. This is rather a synoptic way of stating a pentapod argument which, as we know from the context, runs as follows: 'sound is non-eternal because of being originated; what is originated is non-eternal' and so on. The intended premise of Arthāpatti is 'what is originated is non-eternal'. This example, however, creates some complications for the correct interpretation of Arthāpatti. In the previous example both the premise and the conclusion are compound propositions. But in this case they are categorical propositions. In fact this case belongs to the variety of immediate inference known as contraposition in western logic. Thus it is obvious that the premise of an Arthāpatti is not necessarily a compound proposition. It may also be a categorical proposition, presumably as long as there is only one premise.

The next question is whether, in this second case, the conclusion is related by way of 'opposition' to the premise. There is no question of there being any opposition between the constituent propositions, as there was in the previous case. There is, however, a kind of 'opposition' between the constituent terms in this case. 'Originated' may be said to be 'opposed' to 'non-originated' and 'eternal', to 'non-eternal'. Vātsyāyana has said that 'the positive is the opposite of the negative'. He has not specified that 'opposition' may take place between propositions. Thus it seems that 'opposition' is the relation between not only a proposition and its negation but also between a term and its negation.

Hence it may be said that there is a relation of 'opposition' between the premise and conclusion in both of the available specimens of Arthāpatti. There is no other specimen available. It is possible, therefore, though not so stated in the definition of Arthāpatti, that the premise and the conclusion are to be related by way of 'opposition'. This would certainly amount to an important modification in the concept of Arthāpatti, and we do not know whether such modification would be proper. It is possible that Vātsyāyana left us with a more general definition of Arthāpatti, because he wanted to accommodate

other cases of inference from a single premise to a single conclusion where the premise and the conclusion are not 'opposed'.

Randle has said: "It will be clear that Vātsyāyana means little more by Arthāpatti than what western formalists call the opposition of propositions and immediate inference. But Vātsyāyana has no doctrine of 'logical opposition' as that embodied in our 'Square of opposition'."[7] Now we have seen that there is one example of Arthāpatti which belongs to contraposition as a variety of immediate inference (though Randle has not referred to this example at all). But there is another example (the only one mentioned by Randle), where both the premise and the conclusion are conditional propositions. Reasoning of this type is not traditionally considered under immediate inference. The only sense in which both the example of Arthāpatti may be characterized as immediate inference is that in both there is exactly one premise and one conclusion. Still the important difference between the case of Arthāpatti dealing with conditional propositions and what are traditionally known as immediate inference should not be overlooked.

Secondly it is true that Vātsyāyana has not supplied us with a 'square of opposition'. But it is not clear

that he should be expected to do so. Some of the implications within the square of opposition hold only because of construing that universal propositions have an existential import. We do not know whether Vātsyāyana construed universal propositions as having an existential import. If he did not, some portions of the square of opposition would be invalid to him. There is reason to believe, however, that some of the implications that hold through 'opposition of propositions' were known to Gautama and Vātsyāyana. They undoubtedly knew that a universal proposition could be falsified by producing one counter example which entitled us to assert the truth of the corresponding particular proposition.

The first example of Arthāpatti was: 'if there is no cloud, there is no rain, so 'if there is cloud, there is rain'. This argument is invalid. Vātsyāyana (notes its invalidity while commenting on aphorism 2.2.3, which) says: "Arthāpatti is invalid because of being inconclusive". Vātsyāyana's comments are: "If there are no clouds, there is no rain': from the meaning of this it is obtained that 'if there are clouds, there is rain'. But even if there are clouds, there is sometimes no rain, so that this Arthāpatti is invalid".[8]

This passage shows that Vātsyāyana knows that an argument is invalid if its premise or premises are true and the conclusion is false. That is why he is pointing out that the preceding argument is invalid because the premise, namely, 'if there are no clouds, there is no rain' is true, but the conclusion, namely, 'if there are clouds, there is rain' is false.

Another important thing in this passage is what it reveals about the nature of conditional propositions. Vātsyāyana has referred to the point of time when there are clouds but there is no rain. This interpretation falsifies the proposition that 'if there are clouds, there is rain'. The only condition in which a material conditional is false is when the antecedent is true and the consequent is false. Vātsyāyana correctly identified that condition, though he has not spelt it out in actual words. Moreover, this interpretation makes the premise 'if there are no clouds, there is no rain' is true. This shows that Vātsyāyana correctly realized that a conditional proposition remains true if the antecedent is false and the consequent true. Vātsyāyana, however, has not explicitly supplied us with the truth table of the material conditional as was done by Megarian-Stoic logicians.

After pointing out that the argument is invalid, Vātsyāyana has gone further to indicate what would be a valid argument in the given circumstances. This is found

in his comments on aphorism 2.3.4 that "(Arthāpatti was thought to be invalid) because of considering as Arthāpatti what is not Arthāpatti (prefer)".[9] Vātsyāyana writes:

That even if there is the cause, there is no effect due to obstructing factors is a causal phenomenon and this is not what is known with certainty by Arthāpatti. What then is known with certainty by it? 'If there is the cause, there is the effect' (should be changed to) 'it is not that there is the effect without there being the cause', this is what is known with certainty by it. It should be remembered that these comments are about the invalid argument. Vātsyāyana is discussing. Vātsyāyana is telling us in a more general way that what should be deduced from 'if there are no clouds, there is no rain' is not 'if there are clouds, there is rain', but 'it is not both that there is rain without there being clouds'. This is a valid argument in the form that 'if not A, not B; therefore, not both B and not A'. Randle has stated the valid form specified by Vātsyāyana as: 'if not A, not B; ergo if B, A'. This is not strictly accurate. 'If B, A' is logically equivalent to 'not both B and not A', but still a different proposition.

One interesting question to ask is why Vātsyāyana switched from the proposition 'if there are no clouds

there is no rain' to the more general proposition 'if there is no cause, there is no effect', making similar appropriate changes with respect to the other propositions involved? There is more than one answer possible, but the answer that first comes to our mind is that Vātsyāyana was not concerned with the particular words 'clouds' and 'rain'. He not only wanted to show that the particular argument 'if there are no clouds, there is no rain; therefore, if there are clouds, there is rain' is fallacious, but also that the very form of the argument is invalid, so that any other argument put into the same form would turn out to be invalid. Hence Vātsyāyana used general concepts. The word Kāraṇa, in the present context, may be interpreted as the ground or the antecedent and kārya, as the consequent. Thus Vātsyāyana's statements amount to saying that the following form of argument is invalid, namely, 'if there is negation of the ground there is negation of the consequent; therefore, if there is the ground, there is the consequent'. When he showed that the argument 'if there are no clouds, there is no rain; therefore, if there are clouds, there is rain', is invalid, he may have produced it as a counter example to the invalid argument form. One important ingredient of a logical theory is the method of proving the invalidity of argument forms by producing a counter example, that is, producing an

argument in the same form, the premise(s) of which is true and the conclusion false. It is likely that Vātsyāyana was aware of this method and applied it to the present case.

It will be clear that Vātsyāyana means little more by Arthāpatti than that western formalists call the opposition of propositions and immediate inference. But Vātsyāyana has no doctrine of 'logical opposition' such as that embodied in our 'square of opposition'. He notes that some supposed 'implications' are in fact not logically necessary; but he does not formulate the precise conditions under which an implication is cogent.[10]

Nevertheless the sūtra already defends Arthāpatti against the charge of being inconclusive; although it denies that it is an independent source of knowledge, and holds that it comes under the head of inference. Neither the sūtra nor the bhāṣya makes it clear just what form implication would take when expressed as Anumāna, inference through a middle term. But the sūtra has no doubts as to the identity of Arthāpatti and Anumāna; for it argues against the objector's inference ("implication is not a source of valid knowledge because it is inconclusive"), that, if implication is invalid, then this inference itself invalid; while, if the inference is

valid, then implication is valid; the meaning of the dilemma being that the validity of inference stands or falls with the validity of implication since in fact there is no distinction between implication and inference. Neither the sūtra nor the bhāṣya recognises the distinction, first made perhaps by some Mīmāṁsaka predecessor of Śabara (unless Śabara himself originated the doctrine) between verbal implication (Śrutārthāpatti) and real implication (Dṛṣṭarthāpatti). Praśaṣtapada cognises it, but regards it as of little importance.

All implication is within a system and therefore relational. Vātsyāyana takes his example from the causal relation, Śabara takes his from spatial relations (if a man is not here, he is elsewhere), while the later Mimāṁsaka adds one based on time (if not now, at another time). All three illustrations can easily present themselves as a mere opposition between positive and negative - between this and not this; for the reason that a system is a whole of mutually exclusive parts, and always be expressed disjunctively. This aspect of the relational argument is present to Vātsyāyana when he characterises the argument as based on pratyanikabhāva, or opposition: and when he says the positive is the pratyanika of negative he lapses into a 'formal' view of 'implication'. Śabara and his school, partly atleast, avoid this tendency to

a formal account, because they think of the system of facts as forcing upon the mind a point of view which is not merely the negative of the impossible supposition but a positive conception in itself. Nevertheless they do not succeed in realizing the constructive or 'synthetic' character which is the mark of a genuine Arthāpatti - as exemplified for example in geometrical construction, in which new positive truth arises by implication in the concrete character of a system.[11]

Praśastapāda does not seem to accept Vātsyāyana's account of Arthāpatti; and he draws the distinction, which Vātsyāyana does not draw, between Dṛṣṭārthāpatti and Śrutārthāpatti. His actual words, however, does not enable us to determine what view of Arthāpatti he has in mind. But, for reasons suggested below, it is probable that Śridhara interprets him correctly.[12]

"Implication from an object of experience is no more than inference by opposition: implication from what is heard is inference from the inferred".

Śridhara glosses Virodhy-ānumāna by: "a thing which when contradicted by some other means of knowledge is inseparably connected with another thing is a 'Mark by contradiction'". Absence from the house, when contradicted by knowledge that the man is alive, is inseparably connected

with, and therefore is mark of, being outside the house. Śridhara takes the reference to be to some such view of Arthāpatti as Śabara's.

It is a case of inference, because it clearly rests on avinābhāva, universal connection. "There is a middle term (i.e. the process is inference) just so far as there is a rule of necessary connection", says Śridhara: "If absence in the house caused the thought of presence outside simply through the impossibility of the former alternative (anupapattimātreṇa), it would not be a middle term based on a universal rule (niyamahetu) and therefore the absence from the house might lead to the thought of something else too". (It would not point to any definite alternative). The Mīmāṁsaka says that knowledge of absence from the house generates the idea of something else in order to make itself possible - and it is not possible on any other hypothesis except that of the man's being outside. But on what authority does he affirm that a living man's absence from the house is only possible in case he is outside? On the authority of the experience that a finite substance if it is in one place is not anywhere else. But then the realisation of the possibility of the man's absence from the house has as its condition a positive connection (anvaya), and so is inferential, since its arising depends on a universal connection. The form of

the argument would be: Devadatta is out because being alive he is not seen in the house; like me'.[13]

There is no question that the process is 'inferential' in the sense of depending on a universal. The question is whether it can be 'reduced to syllogistic form': and since the Indian syllogism is in essence inference from example this really amounts to asking whether Arthāpatti can be represented (fairly) as inference from example: and the obvious irrelevance of the example in Śridhara's attempt at reduction to syllogistic form indicates that it cannot be done. Arthāpatti is not inference from examples - Western Logic meets with similar difficulties in attempting to reduce relational arguments to terms of our syllogism: We cannot easily construct a premise stating explicitly the universal from which the conclusion could be supposed to be 'deduced'; and the premises as they stand present a quaternio terminorum: as for instance in such arguments as 'A is to the right of B, and B to the right of C, therefore A is to the right of C'. Indian logic in its doctrine of Arthāpatti is facing this same problem, though from a different angle. The Indian syllogism from example does not differ in essentials from the Western syllogism. Both are subsumptive. And the fact is that there are important classes of inference which are not subsumptive, and therefore refuse reduction to

syllogistic form, or to Anumāna. Indian logic gives a few illustrations of such arguments under the rubric of Arthāpatti! The treatment of the topic is inadequate; but it is still a valuable suggestion.

Praśastapāda's meaning in calling verbal implication Anumitanumāna is merely that verbal testimony is itself (on his showing) inference. The fact derives by 'implication' from words are arrived at inferentially: therefore the 'implication' got from such facts is 'inference from the inferred'.

B. ARTHAPATTI IS A CASE OF SAMĀNYATODRṢTA INFERENCE: UDDYOTAKARA'S VIEW

The whole burden of the Nyāya Vaiśeṣika discussions of Arthāpatti consists in maintaining Devadatta's existence outside the house can be ascertained through inference (Anumāna) and likewise all instances of Arthāpatti can be satisfactorily explained as cases of inference.[14] It argues that neither the instrumental cause of Arthāpatti (anupapatti) differs from that of inference nor the knowledge resulting therefrom differs from inferential knowledge and hence there is no necessity to recognise Arthāpatti as a distinct pramāṇa. The Nyāya-vaiśeṣika system then proceeds to show how Arthāpatti does not differ from Anumāna.[15] It should however be pointed out here that

neither the Nyāya Sūtra nor its commentaries, Nyāya Bhāṣya and Nyāyavārtika take note of the definition and illustration of Arthāpatti as given by Mīmāṁsikas. The knowledge of the fact that "when there is cloud there is rain" does not result from the awareness of the conflict between two well ascertained facts. Arthāpatti according to the Mīmāṁsakās is the presumption of a fact that serves to remove the inconsistency or resolve the contradiction between two well-settled facts. Therefore what passes for Arthāpatti according to the authors of the Nyāyabhāṣya and Nyāya vārtika is not Arthāpatti as understood by the celebrated Mīmāṁsakās. It is only a semantic implication.

It is clear from our foregoing rather detailed account of Gautama's and Vātsyāyana's account of Arthāpatti, that neither the Sūtrakāra nor the Bhāṣyakāra specifies the particular type of Anumāṇa under which they wanted to include Arthāpatti. That is, neither the sūtra nor the Bhāṣya makes it clear just what form implication would take when expressed as Anumāṇa, inference through a middle term. But the sūtra has no doubts as to the identity of Arthāpatti and Anumāṇa; for it argues against the objector's inference ("implication is not a source of valid knowledge because it is inconclusive"), that, if implication is invalid, then this inference is itself invalid; while, if the inference is valid, then implication is

valid: the meaning of the dilemma being that the validity of inference stands or falls with the validity of implication - since in fact there is no distinction between implication and inference.

From what the Nyāya Bhāṣya has said it is obvious that it brings Arthāpatti under anvayavyāpti or positive invariable concomitance. Praśastapāda and his commentator Śridhara openly state that Arthāpatti is the same as anvayavyāpti.[16] Vācaspati Miśra argues for the inclusion of Arthāpatti in Anumāṇa it is clear that he considers it as anvayavyāpti.[17]

Among the Naiyāyikās it is Vācaspati Miśra who for the first time notices and critically examines the definition and illustration of the Mīmāṁsakas Śabara and Kumārila. Udayanacārya ingeniously tries to reduce anupapatti to both Anvaya and Vyatireka concomitances and Virodha to Anvaya concomitance and urges that, in either way, Arthāpatti is not different from inference.[18] Vācaspati and Udayana direct their talent in detecting the presence of anvayavyāpti in the Arthāpatti example of the Mīmāṁsakās, when an existing entity does not exist at one place then it must exist at another place, i.e. non-existent at one place is invariable concomitant with existence at another place in the case of a positive entity ascertained to be existing. This invariable concomitance can

be easily apprehended by anybody with reference to his own body. "When I am not present in one portion of the house, I am present in the other portion of the house". In the given example Devadatta does not exist in his house and therefore it follows that he exists outside. This is an unambiguous case of inference.

Uddyotakāra the author of Nyāyavārthika observes that Arthāpatti as advocated by the Prabhākara school of Mīmāṁsa as consisting in the resolution of doubt belongs to the Sāmānyatodṛaṣṭa variety of Anumāna.[19]

Those who hold that Arthāpatti is produced by doubt argues thus: Through astrology or otherwise it is decidedly known that Devadatta lives for hundred years; but he is not seen in the house; this absence produces doubt both as regards his hundred-year life and his living outside; and to remove that doubt and to establish and ratify his hundred years of his existence outside is postulated. In this inference Devadatta's living is doubted; from doubt we cannot get vyāpti; and as there is no vyāpti this inference cannot be syllogism. It may be objected that if doubt can lead to postulation then, when we see something in the dark and doubt it whether it is a piller or a man, even this doubt must lead to postulation. The answer is that not all doubts ends in postulation; only that

produced by the factors above mentioned leads to it. That is, one alternative like the hundred year life of Devadatta must be an absolute truth, and the other like the expectation of him in the house and not outside must be a supposition; when Devadatta is not perceived in his house there will be conflict between our expectation and our idea of his hundred year life, and the conflict will end in doubt. Such a doubt will lead us to postulation and not anyother. To this again the objection is: If the two alternatives are equally strong, then where is the scope for postulation? If there is a scope, why postulate his existence outside and not his death in the house? If, on the other hand, it is said that the two alternatives are not equally strong, then there is no place for doubt. An ordinary vyatireki syllogism can establish the conclusion. Its form would be: Devadatta lives outside, because he is living but not found in his house; whoever is not so cannot be living and yet be absent from the house. Further, absence from the house cannot produce both doubt and postulation, for what process doubt cannot itself lead to postulation. To this objection also the Mīmāṁsaka is ready with an answer: The two alternatives are contradictories, which cannot both be true and both be false and of which one atleast must be true. If we postulate Devadatta's existence outside then one alternative, his

absence outside and existence inside, is negated but not his hundred year life. But if we postulate his death in the house then both his absence outside and his hundred year life will be negated - which is absurd. (We should note that, because of Devadatta's absence from the house, his hundred year life on the one hand and his absence outside and existence in the house on the other become contradiction as only one of them can be true).

This kind of Arthāpatti in which doubt is karaṇa the Naiyāyikas try to include in their sāmānyatodṛshṭa syllogisms. Gautama's classification of inference is contained in the solitary aphorism: 'Inference is of three kinds, pūrvavat, śeṣavat and sāmānyatodṛṣṭa.[20] In the Sāṁkhya kārika of Īśwarakṛṣna too, it is pointedly stated that inference is of three kinds.[21] In Nyāya Sūtra, three examples of inference are given.[22] They are: the inference of past rain from a swelling river, the inference of future rain from the carrying of eggs by ants, and the inference of the presence of a peacock from its sound. These examples are intended by Gautama to correspond respectively to pūrvavat, śeṣavat, and sāmānyatodṛṣṭa kinds of inference: For Sāmānyatodṛṣṭa Vātsyāyana adduces the example of the movement of the Sun. The cause for the Sun's change of place is motion. When we see the Sun occupying a certain place in the morning and other times during the day and

altogether a different place in the evening, we can infer that it is his motion that is the cause of this.

To the above argument of the Mīmāṁsakās the Naiyāyika answers: Of the two alternatives one is certainly to be denied. The argument that Devadatta's death negates both alternatives while his existence outside negates only one, is only a tarka, which is another way of saying: If Devadatta were dead his age would not have been hundred years of life. Withholding of such tarka and sāmānyatodṛshṭa syllogism negates one of the alternatives by having in view the idea of death, which would be true if Devadatta does not exist outside. To this the Mīmāṁsaka replies: If his existence outside is not known at all, then living Devadatta must be taken to be present in the house only. And if the result of our syllogism is to be the negation of one of the opposition alternatives, it would be impossible without knowing beforehand Devadatta's existence outside, which must have been possible only through Arthāpatti.[23]

It is somewhat difficult to follow this argument. However, it may be expressed thus. Is the existence outside to be inferred from the absence in the house or is the absence in the house to be inferred from the existence outside? First, we start with the idea that Devadatta is in his house; we do not find him and suspect that he

is dead. But his death conflicts with his hundred year life guaranteed by astrology. So his death has to be denied. But what is the basis of the negation? Only his existence outside. Hence without the knowledge of his existence outside one of the alternatives cannot be negated. But how to obtain that knowledge? Can it be obtained simply from the absence of Devadatta from the house? No. For Devadatta might have been dead also. So unless our doubt that Devadatta is dead is removed, we cannot infer his existence outside; but that doubt is removed only when we know his existence outside. The Mīmāṁsaka says that Devadatta's existence outside is known through Arthāpatti and this knowledge negates the possibility of his death; and he contends that the Naiyāyikas cannot say this because, according to him, Devadatta's existence outside cannot be known unless it is decided that he is not dead. Indeed, the Naiyāyikas says that his decision is obtained through tarka, and that ordinary syllogism assisted by this tarka gives the conclusion that Devadatta exists outside.

His sāmānyatodṛshṭa syllogism would be of the following form: Devadatta is either dead or living, because he is an organism like me. Now tarka comes to help this syllogism: If he were dead his age would not have been a hundred years of life and both the alternatives referred to above

would be denied. Then the conclusion is: Devadatta who must be either living or dead, when he cannot be dead, must be living outside. The Naiyāyikas contend that tarka does not assist doubt because it would not be economical (lāghava) to assume that it assists the sāmānyatodṛshṭa syllogism rather that doubt, for doubt is not a pramāṇa.[24]

Some of the later Mīmāṁsakās therefore hold that Arthāpatti is produced not by doubt, but by incompatibility, and that it is a postulate to remove that incompatibility. Devadatta is living; he is absent from his house; his absence is incompatible with his long life; to remove the incompatibility his existence outside has to be postulated. One may here try to frame a syllogism. Devadatta exists outside, because though absent from the house he is living, just as a pot which is not destroyed but absent from the house must be found elsewhere; or because any existent thing, if not found in one place, must be found in another, just as I, standing in the middle of the room, or absent from its corner. But this does not refute Arthāpatti as a separate form of inference, because our knowledge of Devadatta's existence outside can be obtained even when we are ignorant of the concomitance between the middle and the major terms. Sāmānyatodṛsṭavyapti cannot be accepted because there can be no vyāpti of that type.

The Naiyāyika says: What is anupapatti or incompatibility? Incompatibility exists when there is no absence of hetu wherever there is absence of sādhya. Hence there is no need of a separate pramāṇa called Arthāpatti. The Mīmāṁsakas reply: Syllogism is possible only when we have vyāptapakshadharmatājñana, that is when we are able to subsume the minor term under the middle. But here there is no major premise and so no middle term. It may be said that with the help of the vyatirakvyāpti the minor can give the conclusion. But the Naiyāyikas themselves say that Vyatirekavyāpti is of no use in a kevalānyayi inference, and instead of having so many forms of vyāpti, it would be economical (lāghava) to have only one which asserts the direct concomitance (sādhyovyāpyatvam) of hetu and sādhya. Moreover, the absence of Devadatta is found in the room, and how can the hetu of his presence be elsewhere. For the rule is that the hetu and the sādhya should have the same locus (samāṇadhikaraṇya). It may be said that this objection holds only if the room is the anuyogi or the locus of the absence is taken as the hetu, but not if the pratiyogi or the counterpart of the absence is the hetu. The pratiyogi here is Devadatta, and he exists elsewhere. Thus he is the sāmānādhikaraṇa of his absence and elsewhere existence. But Devadatta is not perceived, while his absence is perceived; and so liṅgajñāna or knowledge of the mark is not possible

and there would be no occasion for a syllogism. Further, the Naiyāyikās insist upon tṛtīyaliṅgaparāmarśa or the perception of the mark for a third time. But it is impossible unless Devadatta is perceived.

But the Naiyāyikās say that even tṛtīyaliṅgaparāmarśa is possible through memory. As a matter of fact, the absence of Devadatta in the house is related to both the house and Devadatta. Hence Devadatta is brought before mind, which can treat him as the pakṣa having a vyāptiviśiṣṭaliṅga, that is, as the minor subsumed under the middle. Hence even the Arthāpatti produced by anupapatti is only a syllogism of the sāmānyatodṛshta type.

C. ARTHĀPATTI IS NOT DISTINCT FROM ANUMĀNA : JAYANTA BHAṬṬA 'S VIEW :

Jayanta in his Nyāya mañjari specifically rejects Sambhava, Aithiya, Arthāpatti and Abhāva as distinct sources of knowledge. Jayanta follows the conventional Nyāya line in rejecting presumption as a source of knowledge. Though his predecessors have also dealt with the problem, his detailed, comprehensive and synthetic treatment of Arthāpatti in Nyayamañjarī seems to have surpassed all former accounts. He has vehemently argued for the superiority of the Nyāya exposition against that of the Mīmāṁsā. It is perhaps for the first and the last time that a triangular conflict regarding Arthāpatti has been brought to

surface and resolved with ingenious and erudite argumentation. At the time of Jayanta, the main exponents of Arthāpatti were the Mīmāṁsakās. Thus Jayanta has to analyse this problem against the views of the Mīmāṁsakās. Though his conclusions are conventional, his presentation is novel and unique. He refers to the conflict subsisting between both the schools of Mīmāṁsā. Though at times he seems sympathetic towards the Prābhākaras, ultimately he refutes both the schools contending that since Arthāpatti is reducible to inference, it is not a distinct way of knowing.

Jayanta argues that presumption is identical with inference.[25] He points out that if two things are not connected by a tie of inseparable relation then one cannot throw light on the other. We cannot know another object just on the basis of the knowledge of an object. Again, the relation of vyāpti may exist between the two objects but if we do not know it to be such then we cannot infer the other, by just knowing one of the two. This remark of Jayanta reminds us of the two conditions of valid inference, constitutive and epistemic that Prof. L.S.Stebbing points out in her 'Introduction to modern Logic'.[26] For instance a new born baby is not equipped with such a complicative process of knowledge. In all cases of presumption it is not possible to know the specific relation existing

between the two objects. But it is equally true that they are known to be related in a general way. Jayanta also calls attention to the Mīmāṁsaka position that in the case of Arthāpatti that it (the given) cannot be proved without this (the assumption) is an instance vyatireka vyāpti. The vyatireka vyāpti, being known, also leads to anvaya vyāpti that if it is assumed that it will be explained. Anvaya vyāpti and vyatireka vyāpti are the properties of the hetu. Hence, Jayanta argues, Arthāpatti is non-else than inference.

Bhaṭṭa's conception of Arthāpatti is identical with inference. Jayanta proceeds to prove that the so called instances of Arthāpatti which are based upon Anupalabdhi or non-apprehension and śabda or verbal testimony which are advanced by the Mīmāṁsakās to illustrate the existence of some super sensuous power of an object, are really included in inference as the instances are really pointless since the super sensuous object does not exist.[27] According to Jayanta, the examples of Arthāpatti based upon Anupalabdhi, are the really illustrations of inference.[28] The absence of a living person from his house constitutes a probans which indicates his existence outside his house. The absence of living Caitra from his house is the subject of inference. His existence outside his house is the probandum. The state of his being absent from the house is the probans. On previous occasions we have found that

whenever a living person has been absent from his house he has been present outside the place of his residence. A similar instance may be cited. Smoke co-exists with fire because it possesses smoke. Therefore, it has been a vain feat of wordy warfare to find fault with the reason and to refute the view that the proposition that living Caitra is present outside his house follows as a conclusion.

To the Mīmāṁsaka contention that Arthāpatti is an instance of Anumāna then Arthāpatti should not be a distinct source of the knowledge of new objects. Such contention according to Jayanta is untenable. He asks which object is grasped by Arthāpatti, according to Mīmāṁsakās? Is it unqualified existence of Caitra? or Is it Caitra staying outside the house? Caitra's existence pure and simple has been known from the predictions of astrology. Hence, it is not a new object. According to Jayanta the object to be known by Arthāpatti is Caitra's existence somewhere outside his house. The proof which generates knowledge of his absence from his house opposes the appearance of such knowledge as reveals his presence at home. But it throws light neither on his presence nor on his absence outside.

When we know that a living person is absent from his house then we erringly infer that he is, existing outside the house. Moreover, the knowledge of the absence of a living person from his house cannot be equated with

the knowledge of his presence outside his house. The fact that a man is living is one thing but, on the other hand, the fact that he is staying outside his house. From the knowledge of the first object we pass on to the knowledge of the second object but these two pieces of knowledge are not identical. Our knowledge of smoke on the locus of fire is not the same as is the knowledge of fire. Smoke is different from fire. In this case, too, the absence of a living person from his house is distinct from his presence outside his house. The hill and fire are two known objects. We infer only their relation which is a novel object. Similarly, here too, the connection of the living person with the place outside his house is only inferred since it is such an object as was not known before.

Similarly Jayanta considers Arthāpatti, based upon verbal testimony, is a weak one and is not different from that of inference. It is merely absurd to presume the part of sentence since the meaning of the absent part can be inferred from its effect which acts as a real probans. Just as we infer fire, the cause of smoke, on a mountain seeing smoke, the effect of fire, in its cave, so also we may infer the taking meal, the essential condition corpulence, noticing the fatness of a person. We observe no difference in these two instances. From our repeated observations we have learnt that corpulence is the effect of the eating of food. In some cases the probance

is perceived but in other cases it may be known by means of Śabda. There is no great gulf of difference fixed between these two marks, acquired by perception and verbal testimony.

The Mīmāṁsakās contend that an incomplete sentence cannot convey its meaning. They do not mean to say that the sentence that corpulent Devadatta does not take his meal during the day fails to convey that he is corpulent but simply intends to suggest that the knowledge of his corpulence is conditional and that it involves a reference to its condition which is not given there. The hypothetical knowledge, being incomplete, is not a true one. Hence the source of this knowledge, i.e. the sentence, is not a means of true knowledge. Hence, the sentence in question should be made complete. Jayanta considers this Mīmāṁsa conception also is untenable. He questions which one of the following involves reference to something else. Is it a word or its meaning or its knowledge? If it is held that a word involves a reference then it may be answered that as no inarticulate sound involves a reference so a word which does not refer to a meaning involves no reference to some other word. If it is further held that a word which refers to its meaning also involves a reference to some other word then the hypothesis requires further clarification. The upholders of the hypothesis

should agree to the point that the meaning of a word also involves a reference to some other objects. If it is admitted that the meaning of a word involves a reference to some other object then the hypothesis that the presumption of a clause is necessary becomes superfluous. Now, the Mīmāṁsakās may uphold the third alternative and assert that consciousness involves a reference. If this is Mīmāṁsakās stand, then Jayanta points out that all forms of consciousness do not involve a reference to other objects. Jayanta says that Nyāya admits that the knowledge of the meaning of a word involves a reference to other objects over and above its own objects because its object involves a reference to them. But Nyāya also subscribes to the view that the auditory perception of a sound does not involve such a reference since sound itself, its object, does not involve a reference to other objects. Hence, in order to establish the referential character of meaning one should admit the referential character of an object.

The Mīmāṁsakas further presume a clause in order to establish the knowledge of some additional objects. Hence, it will be a similar course to be adopted by us if we directly assume the additional objects themselves. What benefit, Jayanta asks, do we derive, having recourse to the additonal step?

In this connection the Mīmāṁsakās raise an objection that what is assumed does not follow from Vedas. Jayanta tries to meet this objection that even if we assume for the sake of an argument that a clause or a sentence is presumed than the meaning of a clause or of the sentence is not conveyed by the same since either of them is non-Vedic. Jayanta also counts the possible contention of the Mīmāṁsaka the Vedic texts is of two fold character viz., the first type is heard, i.e., given and the second one is inferred and that what is presumed pertains to Vedic texts.[29] His counter argument is that the subject-matter of the Vedic text admits of two kinds, viz., the first one is directly given by the Vedic texts and the second one is suggested by the first one. Thus, the suggested matter is also Vedic. In that case, why should we subscribe to a gratuitous assumption in the shape of a clause or of a sentence? Hence, the new matter, i.e. the subject matter, which is derived from the direct vedic text will be treated also as vedic. Therefore, we can confidently assert that the hypothesis of presumption based upon the verbal testimony is in no way more satisfactory than that of the Naiyāyikas.[30]

Criticism of Śrutārthāpatti

Neither the Nyāya Sūtras nor the Nyāya Bhāṣya recognises the distinction first made, by the Mīmāṁsakās between

Śrutārthāpatti and Dṛuṣṭārthāpatti. Praśastapāda of Vaiśeṣika cognises it, but regards it as of little or no importance. Jayanta who takes the Mīmāṁsaka distinction seriously specifically directs his attack on Śrutārthāpatti as non-different from inference. The following are his arguments to prove his thesis and to refute the stand point of the Bhaṭṭa Mīmāṁsa.[31]

1) The example of the so called presumption, based upon the verbal testimony, is a weak one, since Śrutārthāpatti is not different from an inference. It is really absurd to presume the part of a sentence; since the meaning of the absent part also can be inferred from its effect, which serves the purpose of a genuine mark. As fire is inferred from smoke, similarly one can infer the taking of meal as the mark (essential condition) of flabbiness on the basis of noticing the great bulk of a person. There is no difference in these two instances.

2) From our repeated observations we have learnt that flabbiness is the effect of the taking of meal. In some cases the mark is perceived but in other cases it may be known by means of the verbal knowledge. As there is no great gulf of difference between these two types of mark, there is no difference between the Śrutārthāpatti and inference.

3) The Bhāṭṭa Mīmāṁsakās contend that an incomplete sentence cannot convey its full meaning. They do not mean to say that the sentence that 'Flabby Devadatta does not take his meal during the day', fails to convey that he is flabby but what they simply intend to suggest is that the knowledge that "he is flabby" requires a complement, i.e. 'he takes food at night' which is not given in the basic form of the sentence. So what the Bhāṭṭās do is, that they propose to presume the complementary portion of the sentence and call it '<u>Śrutārthāpatti</u>. But Jayanta puts them in a dilemma by asking them as to which of the constituents of the given sentence require a complement? Is it a 1) word or 2) its meaning or 3) its knowledge?

1) If the Bhaṭṭas hold that a word requires complement, then it may be said that as no inarticulate sound requires a complement, so no word which does not refer to a meaning requires a complement.

2) If it is the contention of the Bhāṭṭās that the meaning of the given sentence requires a complement, then let its requirement be fulfilled by having a complementary meaning. And in that case, there is no point in presuming a portion of the sentence.

3) If the knowledge brought about by the given part of the sentence is held as requiring a complement, in

such a case even it is the object which is required to make the already acquired knowledge comprehensive, and for that purpose the simple course to be adopted is that the additional objects themselves should serve as complements. And in this situation, the presumption of a part of the given sentence becomes superfluous.

Jayanta is aware of the objections that the Bhāṭṭās could raise against his refutation of their preposition. So at this juncture also he reproduces and then rejects their counter arguments. 1) The Bhāṭṭās state that the presumption of the object and not of the reference themselves would render the Vedic sentence non-vedic. But Jayanta holds that if the sentence is presumed, the meaning would become non-vedic with the same logic, since the presumed portion of the sentence is non-vedic is weak shelter. 2) Moreover, the directly given sentence and the subsequent portion indirectly assumed from it, both constitute a vedic text according to the Mīmāṁsakās, then with the same logic the given and suggested meaning would also be regarded as Vedic. And in the case, there is hardly any propriety in the gratuitous presumption of a clause or a sentence.

Jayanta further holds that on the basis of the above mentioned facts, the arguments of those also automatically stand refuted, who hold that 'Flabbiness' as in the given

example is sensuous and 'taking food at night' is supersensuous and the relation of concomitance holding between them cannot be ascertained. Jayanta seems to propound that if one portion of a logical whole is ascertained, the remaining part could be inferred, and the injection of supersensuous colouring to the silent part is totally unwarranted.

On reviewing the aforesaid account of Śrutārthāpatti we arrive at the conclusion that 'Śrutārthāpatti' is not a distinct means of knowledge. The arguments are not a distinct means of knowledge. The arguments advanced by the Bhāṭṭās are not based on sound footing.

As regards Dṛṣṭārthāpatti, Udayana puts forward the curious argument that, though the existence of a living person outside his home is not in conflict with his absence from home, being alive and being absent from home are in conflict with each other, and consequently that the presumption of a living person's being outside his home does not really deserve to be so called, but should be regarded as an opposing inference (Virodhi Anumāna) Śridhara and Śamkara Miśra, on the other hand, offers more plausible argument in favour of the view that Dṛṣṭārthāpatti is identical with inference. According to them, the presumption that Devadatta lives outside his home

is really inferential knowledge in as much as it is dependent on the invariable concomitance (vyāpti) between a probans, namely, the non-existence at home of Devadatta as a living person, and a probandum, namely, his existence outside his home. It is to be noted, however, that the invariable concomitance referred to here is positive (anvayi) and not negative (vyatireki). That being so, the Vedānta seems to lose its point in so far as it holds that presumption may be shown to involve negative concomitance, but never the concomitance of the positive kind.

So far we have referred to Jayanta's criticism of the Bhaṭṭa school of Pūrvamimāṁsa. We shall now turn our attention to his refutation of Prābhākara's defence of Arthāpatti. Prābhākara and his followers adopt a different line of argument to distinguish Arthāpatti from Anumāna.[32] In case of Anumāna the hetu is not dependable if the sādhya does not exist on the locus of the hetu in question. For instance smoke does not logically exist on the subject of inference (pakṣa) if fire does exist there. Prābhākara thinks that reverse is the relation existing between the implier and implied in the case of Arthāpatti. Taking a familiar example the Prābhākarās point out that it cannot be proved that a living person staying outside his house if he is not absent from his house. When a living man goes out of his house he stays out of the house. The Prābhākarās also explain the extract of

Śabara's commentary to fit it into with this new hypothesis. The object which is known to us either through Śabda or through any other pramāṇa causes us to presume other objects. As such an object is the only source of postulation, so in the absence of it such a presumption becomes impossible. Moreover if we have no opportunity of postulating an object than the object which is postulated cannot appear on the scene. Postulation reveals an object which is assumed. Hence, postulation is different from inference. This is the contention of the Prābhākara school of Pūrva Mīmāṁsa.

The Mīmāṁsakā's distinction between inference and implication referred to above, Jayanta considers as only verbal and is not based upon new facts. If the object to known is not known before, Jayanta asks, then how can one know that it will appear to us to be a puzzle without having such and such a condition? On the other hand if one already possess the previous knowledge of the object then what is the merit of exposing the puzzling character? This is because, Jayanta answers, we have definitely known the object which we are now trying to know in this context. Jayanta anticipates Prābhākara contention that we have already known not merely the object but also the puzzling character without assuming such and such condition would not such a contention asks Jayanta

would amount to the position that the knowledge of Vyāpti precedes presumption? If Prābhākara says that the presence of a living person outside is impossible without his absence from house then they should also concede that if a living person is absent from his house then he is surely present outside his house. Jayanta asks what is the nature of the impossibility? If it is logically impossible then the so called Arthāpatti is no better than Anumāna since the thought process of the latter essentially moves in a similar fashion. If the hetu is known before then and then only the sādhya is known afterwards and the hetu is not previously known then the sādhya cannot be known.

Jayanta feels the statement that the absence of a living person from his house objectively determines his presence outside, is wrong.[33] If absence from his house is an established fact then its effect is subsequently produced by it since a cause is invariably antecedent to its effect. Such a cause alone is productive of its effect. Jayanta observes here that any attempt to establish causal connection between the absence of a person from his house and his existence outside at the next movement would result in an absurd situation viz., at the very movement of the absence, he neither stays in nor outside his house. Hence, the causal relation does not

hold between one's absence from house and his presence outside. All these discussion according to Jayanta is fruitless since it does not in any way improve Prābhākara thesis that Arthāpatti is an independent pramāṇa.

Jayanta also refers to some followers of the Prābhākara school who hold that Arthāpatti is distinct from inference since in it the implier points to and conditions the implied. In the case of inference smoke is only the mark of fire but in the case of Arthāpatti the absence from house both indicates and produces his presence outside. Jayanta also refutes this thesis. He observes that the Mīmāṁsakas themselves are not sure whether or not an invariable relation holds between the implier and the implied and consequently have put forward another example to prove their thesis viz., when a living person is at home he is not out. Jayanta feels that if the Mīmāṁsakas felt that this is the better example they should have given at outset. But this example does not illustrate Arthāpatti based Anupalabdhi or non-perception.

D. ARTHĀPATTI INDISTINGUISHABLE FROM INFERENCE BASED ON KEVALAVYATIREKI VYĀPTI

Of all the attempts of the Nyāya-vaiśeṣika to reduce Arthāpatti to Anumāna the most significant one is to identify Arthāpatti with inference based on kevala-vyatireka-vyāpti or the purely negative concomitance. Udayana,

Jayanta, Gangeśa and a host of Nyāya-vaiśeṣika philosophers of later times contend that Arthāpatti is identical with Anumāna of the Kevalavyatireki type. Udayana affirms that what is Arthāpatti according to Mīmāṁsakas is none other than vyatireka concomitance; the difference is only in name.[34] Before we proceed to consider this contention of the Nyāya-vaiśeṣika philosophers, we had better make a brief note of their classification of inference anvaya vyatireki, kevalānvayi and kevala vyatireki.

In view of the methods of proving the Vyāpti, syllogisms have been classified into anvaya-vyatireki, kelalanvayi and kevla vyatireki. The methods applied in inductively establishing vyāpti are anvaya (agreement in presence) and vyatireki (agreement in absence). In the syllogism, wherever there is smoke there is fire, there is smoke in the mountain, therefore the mountain has fire", the truth of the probandum may be tested both through the methods of anvaya (by observing in different cases, that smoke is accompanied by fire), and also through the method of vyatireka (by observing in different negative cases that where there is no fire there is no smoke).[35] Application of these two tests is possible because, fire, the probandum here is of a nature that allows the observation by both anvaya and vyatireka instances.

The reason which has concomitance with the object to be inferred both by presence and absence, is called 'the positive and negative reason' (anvayavyatireki-hetu). Putting P for the probandum, S for the subject and M for the reason, the two forms of the same syllogism may be represented as follows.[36]

Anvaya	Vyatireka
All M is P	No non-P is M
All S is M	All S is M
∴ All S is P	∴ All S is P

(Here P is such that both 'All M is P' and 'non-P is M' can be ascertained through observation).

Since there is an equality or co-extension between these positive and negative forms, it becomes sufficient to express one aspect only, either the anvaya or the vyatireka. The counterpart of it will necessarily be implied.

But there are certain syllogisms which are founded on the positive concomitance alone or on the negative concomitance alone. A syllogism is called Kevalānvayi when it is founded on a reason which is always positively related to the object to be inferred. For instance the syllogism may be, "whatever is knowable is nameable, the pot is knowable, therefore the pot is nameable. "Here

the reason 'Knowability' is concomitant by presence' with the probandum 'nameability', for it is a fact that whenever there is a knowability there is also nameability. For example, a piece of cloth is knowable and at the same time nameable. But in the mark 'knowability' has no concomitance by absence' with 'nameability'. For there is no such fact which can be truly represented by the statement that whenever there is absence of nameability, there is also the absence of knowability, because 'absence of nameability' and 'absence of knowability' represent no factual characteristic of any real thing, since every real thing is both nameable and knowable.[37]

Symbolically represented the Kevalānvayi syllogism stands thus:

All M is P
All S is M
∴ All S is P

Here non-P is unknown and the absence of M is non-P cannot be ascertained.

Then, there are syllogisms which, although they are based on concomitance by absence, still are not based on concomitance by absence, still are not based on concomitance by presence. Such a syllogism is called Kevala vyatireki anumāṇa. This is illustrated in the example,

"of the five elements none that is not different from other elements has smell. Earth has smell. Therefore earth is different from the other elements".[38] The invariable concomitance which has been utilised here is: 'wherever there is absence of the different from other things'. there is the absence of smell, too. And this amounts to concomitance by absence'. But for the purpose of the present syllogism, no illustration is available for 'concomitance by presence' which could be expressed as "where there is smell, there is 'difference from other things'. *It is true that, earth could serve as an illustration* for such 'concomitance by presence' since it has both smell and difference from other things. Still it cannot do so on the present occasion, because earth is the subject of the conclusion. "Earth is different from other things". If we did offer 'earth' as an illustration for such a concomitance by presence, it would amount to taking for granted what is sought to be proved by the syllogism. Symbolically put the syllogism stands thus:

No non-P is M
All S is M
∴ All S is P

(Here P is such that 'All M is P can never be ascertained through observation).

Uddyotakara, the first known Naiyāyika to speak of Kevala Vyatireki, illustrates the Anumāna in question by using it as a weapon against the exponents of the no-soul theory (nairātmyavādins) like the Carvakas etc. His illustration runs as follows:

The animate body which has life is not without the soul. It is so, it should be without life etc. The thing which is accepted by both the schools without life etc. is seen soulless but this body is not lifeless. Hence it is not soulless.[39]

Uddyotakara specifies also the purpose of the avīta kevala vyatireki by saying that it is intended only to refute the opponent:

The avīta is intended to refute the argument of opponent.

It may be of interest to note that the purpose of the avīta seems more or less the same with the Sāmkhyas also. For, they too employ the vīta-hetus to establish the satkārya theory by saying:

"Effect subsits even prior to the operation of the cause: for that which does not exist cannot be by any means brought into existence: further, only appropriate materials are selected; every thing is not by every means

possible: capable causes produce only that which they are competent to produce : and the effect is not different from the cause".[40]

The effect even prior to its manifestation always exists as a real entity in its cause (which also must always be real entity), as the following considerations will show:

1) Asadakaraṇāt: Nothing can be produced out of a thing which is as non-existent as the barren woman's son.

2) Upādānagrahaṇāt: In producing anything, one has to have recourse to the proper materials out of which only that thing can be produced. We cannot produce curd from water. Only milk can produce it. This shows that the effect has a certain fixed relations to its cause.

3) Śarva Śaṃbhavābhāvāt: It is not possible to press out oil from sand. Oil can be obtained from mustard seed or such other seeds in which it exists. This shows that the effect always is latent form in its cause, otherwise, it would be possible to produce all things from anything.

4) Śaktasya Śakyakaraṇat: It is common knowledge that the effect must be such as is within the power of the cause to create. There must therefore be a relation between the cause and the effect as regards potency also.

5) Karaṇabhāvāt: The cause and its effect have inherent or intrinsic similarity or they may be as non-different as the woven cloth from its cause, viz., the threads.

According to Sāṁkhya, everything has a cause. The cause and its effect always coexist even before the latter becomes known or visible. Nothing happens by chance. Chance according to Sāṁkhya, is a meaningless word needed by us to cover our own ignorance when we cannot ascertain the cause of a thing. As shown in Kārikā III above, the whole universe is a continuous process of change of causes into their effects. The root cause, which itself is causeless, is Avyakta of the next kārika.[41] On the other hand, they use the avita-hetus to refute their opponents, such as the Naiyāyikas who hold to the duality of the material cause and its effect. Their avita syllogism runs as follows:

The cloth is not different from its threads because it contains same quality which is different from other object that would not possess the same quality, for example cow is not having any of the qualities of horse. The cloth contains same quality of threads. Hence it is identical.[42]

However, slowly the idea of kevalavyatireki as a means of Vaidharmya-jñānā asserted itself and hence we find Gangesopādhyāya (C.A.D. 1200) providing an example of kevala vyatireki.

prthivi itarebhya bhidyate, gan dhavattvat

The earth is different from water, fire, air and sky.

Thus the Kevala vyatireki anumāna came to play an important role in the Sāṁkhya, Vaiśeṣika and Nyāya schools. Perhaps it is necessary for the Yoga school also as it has much in common with Sāṁkhyas.

On the other hand the Kevala vyatireki has been summarily rejected by the Mimāṁsakas and the Vedāntins. They accept Arthāpatti as the fifth means of valid knowledge. As we have already pointed out in our exposition of the nature of Arthāpatti. The Mīmāṁsakas say that they need Arthāpatti for certain purposes like assuming a śruti on the basis of a given smṛti, as explained by Kumārila Bhatta;

Smṛtyā śrutiryā parikalpyate smin, etc.

In the "Mīmāṁsā-śāstra" 1) Wherever a śruti is assumed on the ground of another śruti, 2) when a passage is assumed to apply to a definite sacrifice through "power, etc." and 3) where the result, etc. (of a sacrifice) are assumed from outside, - in all these cases we have no conception of any inferential connection.[43]

Similarly Advaitins swear that they require Arthāpatti in order to establish the prapañcamithyātva (illusoriness of the objects) and so on. Besides, both the Mīmāṁsakās and the Vedāntins make use of Arthāpatti also wherever the Naiyāyikas and others employ the Kēvalavyatirēki. The different interpretations and definitions offered by these two opposed schools, then representative concepts and their arguments and counter arguments, since the days of Gautama, Vātsyāyana and Praśastapāda - through the ages of Uddyotakara, Kumārila Bhāṭṭa, Vācaspati Miśra, Udayanācārya, Parthasārathi Miśra, Śrīharsa, Gaṅgeśopādhyāya, Raghunātha Śiromani, Dharmarājādhvarim, Viśvanātha Pañcānana and many others - are well known to scholars and they need not be touched upon here. Thus the Kevala vyatireki divides the Vaidika-darśanas into two conflicting camps - the Sāṁkhyas, the Yogas, the Naiyāyikas and the Vaiśeṣikas, all upholding it on one side, and the Mīmāṁsakās and Vedāntins, rejecting it on the other. Both go to the extent of swearing by their respective anuvyavasāyas of the forms anuminomi and arthāpayāmi (or kalpayāmi). However, on the basis of what we have seen so far, an impartial student may observe this: Both these groups earnestly desire to establish the (itarabheda) uniqueness of the Earth (Pṛthivi); the absence of soullessness of the living body; and both the groups are very much interested in assuming an unknown śruti (a vedic

injunction) on the basis of a smṛti of unknown origin. The thinkers of the first group try to achieve it through the Anumāna alone obviously because the founders of their respective schools do not subscribe to the idea of Arthāpatti as a means of knowledge. At the same time the later writers of these schools themselves have realized that an ordinary Anumāna cannot deliver the goods. And hence they invented a special Anumāna and gave it the name avita or kevala vyatireki. Similarly the teachers of the other group cannot escape noticing the strong Anumāna elements contained in what they call Arthāpatti, in spite of its uniqueness. What they have in their mind is quite discernible when we read Kumārila Bhaṭṭa's concluding observation on the topic:

evaṃ-svabhāvo'pyanumāna-śabadam
labheta ced astu yathepsitaṃ nah

"If you get this nature of the word Anumāna, it is well and good our intention is fulfilled".

A few sūtras of Jaimini also may support this suggestion of ours. For, we have already seen that according to the Mīmāṁsakas etc., the assumption of śruti on the basis of a smṛti of unknown origin is possible only by Arthāpatti, as Kumārila puts it. But atleast in two instances, the sūtrakāra Jaimini calls such as assumption Anumāna only and not Arthāpatti.

api vā kartṛsāmānyāt pramāṇam anumānam syāt and virodhe tvanapekṣam syād asati hy anumānam.[43]

"Not so; by reason of the common author, the non-Veda may also be an authority".

The author gives his own view. Before we see what reply he gives we must determine in what sense he uses certain words Anumāna is used for aśabda śabda is Veda because that alone is the word of God and you are bound to bow down to its authority whether you agree with it or not. The word aśabda is used for smṛti and it is a belief that every dictum laid down in the smṛti is supported by the Vedic authority behind it; if there is none available the presumption is that such authority is lost. This is the meaning of the author by Anumāna or inference. The smṛitī's therefore deriving their authority from the Veda are said by the author as being the work of common impersonal being. Such being the case they are authorities.

Kumārila's Vārttika also in the context runs:

Virodhe tranapekṣam syād asati hy anumīyate.

When there is contradiction it is not to be accepted; when there is none then there is the presumption.

In the preceding adhikaraṇa we have seen that our author says that the smṛties are authorities because they

have the support of the Veda behind them. If we do not find the express Vedic text to support which is lost. Further question that naturally arises is "what are we to do when the smriti text is in contradiction with the express Vedic text?" The reply according to our author is that the smriti is to be rejected in such a case because the presumption as to its validity arises when there is no such contradiction.[45]

However, in the same context he writes a bit earlier:

arthāpattyāpi yatkimcin mūlam ity avagamyate.

It is known that the valid knowledge Arthāpatti has some other sources.

Further, it may also be noted that in the Brahmasūtra-

Śabda iti cet, na; ataḥ prabhavāt, pratyaksanumānābhyam.[46]

The word Anumāna denotes a smṛti that leads to the inference of a śruti. And Śaṃkara too speaks of some subject matter of a smṛti leading to the inference of a śruti:

tat smaryamāṇam rūpam mūlabhūtām śrutim anumāpāyat.[47]

All these would show that even these opponents of Kevalavyatireki find it difficult to think of Arthāpatti unalloyed with the Anumāna element.

Let us now proceed to see how the Naiyāyikas reduce Arthāpatti to an inference of the Vyatireki type. Negative (vyatireka) concomitance is pressed into service when no positive (anvaya) concomitance between the probans (hetu) and the probandum (sādhya) is available. For example, Earth differs from other categories (itarabhinna). This difference from other categories exists only in the earth. If one wants to infer difference from other categories with respect to earth (minor term) he must possess with him, beforehand, a knowledge of that special characteristic, of earth, which is invariably concomitant with "difference from other categories" (itarabhedavyāpya). Since itarabheda and smell are exclusively present only in earth we cannot come across an instance (anvaya dṛṣṭānta) other than earth where we can observe agreement in presence (anvayasahacara) so as to be able to arrive at the positive universal proposition "whatever has smell is different from the rest". But there are instances like water, fire etc., where we can observe at the negative universal proposition. Whatever is not different from others (water, fire etc) has no smell". Difference from other categories, in earth, will have to be inferred only from the negative universal proposition.

None that is not different from earth possesses smell.
Earth has smell.
Therefore earth is different from other categories.

Now it remains for us to see Arthāpatti can be satisfactorily explained as a case of inference of the kevalavyatireki type. It is well known that inference of the kevalavyatireki type starts with a purely negative proposition. It is not indeed an anvayi inference in which the major premise expresses a positive relation of agreement in presence between the middle and the major term, e.g. 'whenever there is fatness, there is eating at night'. On the other hand, it is a vyatireki inference in which the major premise expresses a universal relation between the absence of the major and the absence of the middle. Thus the above example of Arthāpatti may be reduced to the following syllogism:

A man who does not eat at night while fasting by day is not fat;

This man who fasts at day is fat;

∴ This man is not a man who does not eat at night, i.e. he eats at night.

As Arthāpatti may thus be reduced to vyatireki inference, the Naiyāyikas refuse to acknowledge it as a separate source of knowledge.[48] So also the Sāṁkhya philosophers explain Arthāpatti as a form of inference. Taking the second example of Arthāpatti given above, Vācaspati points out that it can be reduced to the following inference;

If a living individual is absent somewhere, he is present elsewhere;

Devadatta who is living absent from home;

∴ He is somewhere outside his home.

Here a man's existence outside his home is inferred from 'his absence from home' as the liṅga or the middle term. There is a relation of vyāpti or universal concomitance between a man's presence somewhere and his absence elsewhere. Everyman finds this to be true in his own case. Hence when we know the one from the other we simply infer its liṅga or universal concomitant, just as we infer fire from smoke.[49]

Now it remains to see whether the Naiyāyikas have suceeded in maintaining Kevala vyatireki as an Anumāna. It appears that they have not succeeded in their efforts. Uddyotakara's Kevala vyatireki Anumāna, quote above, it is to be admitted, smacks of Arthāpatti. And this seems to be indicated also by his own statement that it is meant not for establishing a new truth, but only for refuting opponents who raise objections against the conclusions of a system. The point may become further clear if we examine the question as follows: The form of the Vyatireka-vyāpti (universal pervation of the negatives of two things) is given by the Naiyayikas as: Sādhyābhāva-vyāpākibhutābhāvapratiyogitvam. If the knowledge of this vyāpti

is taken to be the karaṇa or instrument of the kevala-vyatireki inference, then how can the inference be avoided when one entertains the idea that the hetu is vyabhicarita, namely, it is sadhyābhāvavadvṛtti, meaning the hetu (the ground of inference) exists where there is no sādhya (that which is to be inferred)? That is to say that anumiti would be inevitable even when one entertains the above idea. For, the said knowledge of vyapti cannot be obstructed by the above-mentioned knowledge of vyabhicāra. On this score, namely the lack of proper bādhyabādhaka-bhāva (the relationship of the hinderer and hindered), Raghunātha Śiromaṇi has rejected the Kevala vyatireki vyāpti and concludes that a vyapti (concomitance) like sādhyābhāvāvad-avṛttiḥ alone can be the cause of anumiti in all instances and hence the knowledge pṛthivi itarebhyo bhidyate, gandha-vattvāt and the like are to be considered only as Arthā-patti and not Anumiti.

Again positive (anvaya) proposition. "Whichever does not exist in the house exists outside" when transformed into a negative proposition will assume the form "whichever does not exist outside is not non-existent in the house. This amounts to saying "whichever does not exist outside the house exists in the house". Here non-existence outside and the non-existence of non-existence do not occur in one and the same locus i.e. the locus of non-existence

is outside the house and locus of the non-existence of non-existence is in the house. It appears that there is no samānādhikaraṇa and hence no vyāpti.

In the classical example for Kevalvyatireki type of inference both the "non-existence of difference /of difference from other categories" and "non-existence of smell" have the same locus, viz. water etc. In the vyatireki inference one can observe smell in the middle term (pakṣa): but absence in the house cannot be seen in the pakṣa, viz. Devadatta. How could inference originate in the absence of the knowledge of the middle term in the minor term? i.e. when there is no pakṣadharamtā?

The Naiyāyikas have surely overlooked the difference between vyatireka concomitance and Arthāpatti as they have done in the case of anvaya concomitance. The postulation of Devadatta's existence outside does not arise from a knowledge of vyatireka concomitance. The standpoint that Arthāpatti is none other than vyatireka concomitance, therefore, falls to the ground. Under this situation it is left for the Naiyāyikas either to give up their stand or to modify the definition of vyāpti suitably so that it may have an extended application and thereby bring Arthāpatti under its purview.

If, in spite of the fact that knowledge of vyāpti has no function to perform in the emergence of presumption, the Naiyāyikas stick to their position, viz. presumption is the same as the inference of vyatireka type they have to explain the psychology of inference. That inference is by conception a process of reasoning based on an invariable concomitance between the middle and the major terms is evident from the definition of Anumāna as enunciated in the Nyāyabhāṣya.[50] From this definition it is definite that inference starts with anvaya concomitance. But vyatireka vyāpti is the invariable concomitance between the absence of the major term and the absence of the middle term. Invariable concomitance between the absence of the probandum and the absence of the probans is totally irrelevant when the probandum is sought to be established by means of a knowledge of the probans. How can vyatireka vyapti which has no causal connection with inference generate inference? Udayanacarya contends that a vyatireki probans is invariably connected with the probandum (anvayena vyāpta). The apprehension of this anvaya concomitance is effected sometimes by agreement in both presence and absence and sometimes by either.[51] In other words knowledge of the inseparable association (vyatireka sahacara) between the absence of the major and that of the middle term yields the knowledge of the universal concomitance between the middle and the major and generates an inference through the latter.

The stand that Udayana has taken is suicidal. If the knowledge of the vyatireka association too yields knowledge of anvaya concomitance then, according to Udayana, there remains only one vyāpti viz. anvayavyāpti. Vyatirekavyāpti has no place and the term Vyatirekavyāpti will have to be understood in a secondary sense viz. 'that anvayavyāpti which is derived from the knowledge of vyatireka-sahacara.[52] Further it amounts to this-vyāpti is one, anvaya and inference is only two-fold, anvayi and vyatireki. The time-honoured Nyāya division of vyāpti under two heads and of Anumāna under three heads carry no significance: it must be given a goodbye.

Further the statement that vyatireka association yields anvaya concomitance will be meaningful only when there exists atleast one instance, other than the minor term, where the middle term is known to reside. But Kevalavyatireki probans, by definition (pakṣamātravṛtti) abides in only one locus, viz. minor term; consequently no anvaya concomitance is possible. As such to say that vyatireka association yields anvaya concomitance and thereby yields inference is inconsistent with the Nyāya conception of a Kevalavyatireki anumāna.

Jayanta Bhaṭṭa has a different answer to give. According to Jayanta the vyatireka proposition "non-existence in the house is unintelligible without existence outside"

necessarily implies the anvaya proposition "non-existence in the house is intelligible only in the light of existence outside".[53] A kevalavyatireki probans leads to inference only with the assistance of the intermediary anvaya concomitance.

This anvaya concomitance cannot be ascertained with reference to any other locus since the vyatireki probans abides in the minor term only. Therefore it has to be admitted that anvaya is apprehended either by implication or is inferred from vyatireka concomitance as Gangeśa holds.[54] In either way vyatireka concomitance stands in need of anvaya concomitance in order to generate inference and by itself it cannot. When the dependence on anvaya concomitance has become unavoidable the probans loses all its claims to the title Kevala vyatireki; it turns out to be Anvaya vyatireki ultimately.

He further observes that a probans (liṅga) is productive of inference only when it is known to possess both the properties of anvaya and vyatireka. He also admits that the instrumentality of a kevala vyatireki probans consists in its sole dependence on anvaya concomitance.[55]

The stand taken by Jayanta runs contrary to the Nyāya conception of a kevalavyatireki probans. In his anxiety to include presumption under vyatireka inference he abandons

the Nyāya position and unwittingly creeps into the camp of his opponents - the Bhaṭṭas and the Vedāntins who maintain that all vyāpti is only anvaya.

Gangeśa upadhyāya contends that a vyatireki probans is conducive to inferential knowledge "in its capacity of being the counter positive of the non-existence which is pervasive of the non-existence of the probandum".[56]

This answer is not at all convincing for its does not bring out the relation between the probans and the probandum in the Kevala vyatireki type of inference. This is not a fitting answer to the charge that the relation between the absence of the probandum and the absence of the probans is irrelavant when the sādhya is to be established though its relation to the hetu. It is a mere verbal manipulation. In spite of the transformation of the vyatireka proposition the fact continues to remain unaltered. "Being the counter positive of the non-existence which pervades non-existence of the probandum" is materially the same as "the non-existence of the probandum invariably accompanies the non-existence of the probans".[57]

Further this cannot be an adequate definition as the element of sāmānādhikaraṇya, the essential constituent of vyāpti, is glaringly absent. In fact this forms an appropriate definition of Arthāpatti.[58]

Uddyotakara and, on the guideline shown by him, Ramakṛṣṇadhvarin come forward with a different reply.

Inconstancy (vyabhicāra) is opposed to vyāpti and a probans cannot give rise to inference so long as it is known to be inconstant (vyabhicāri). It is logically consistent that an anvayi probans is capable of generating anumiti if and only if its anvaya relation to the probandum is known to be not inconstant (avyabhicāra). Absence of inconstancy is a necessary condition for an anvayi hetu to become operative. It is found that a vyatireki probans also gives rise to anumiti only when its relation (vyatireka sahacāra) to the probandum is free from inconstancy. What makes the anvayi probans operative makes the vyatireki probans also operative. Therefore it is idle to ask how a vyatireki probans can generate anumiti. Ramakṛiṣhnadhvarin sums up the whole argument in a nutshell: Knowledge of vyatireka concomitance generates inferential knowledge by virtue of its being opposed (virodhi) to the knowledge of inconstancy in the hetu (vyabhicārajñana).[59] It follows that knowledge of vyāpti-whether of the anvaya type or of the vyatireka type is productive of inference because of its possession of the common element, viz., opposition towards knowledge of inconstancy (vyabhicāra).

It will be highly enlightening to know that the famous neologician Raghunatha Siromani has to say in this context.

By means of a penetrating analysis of the nature of vyabhicāra and Vyatireka vyapti he demonstrates the utter impossibility of working out any real opposition between the knowledge of the two. One cognition can impede the emergence of another cognition only if both the cognitions bear the same mark of distinction (samānaprakāraka). To witness a specific instance.

The vyatireka concomitance will assume the form, "The non-existence of smoke is not the counter-positive of that non-existence which occurs in the locus of the non-existence of fire" [60] and vyabhicāra assumes the form "smoke is the counter positive of that non-existence which occurs in locus of the non-existence of fire".[61] The mark of distinction of the cognition of the vyatireka concomitance is dhumabhavatva and that of the cognition of vyabhicāra is dhumatva and as such there is no occasion for opposition (virodha) between the cognition of vyatireka concomitance and vyabhicāra.[62]

CONCLUSION

The foregone scrutiny of the Nyāya position makes it plain that the validity of Arthāpatti as a distinct means of knowledge remains unaffected in spite of the unsparing attack from the Nyāya side. The vehement opposition to Arthāpatti is not founded on valid grounds: probably it is based on a firm conviction that the number

of pramāṇas cannot be in excess of the four enumerated by Gautama and rooted on a spirit of unwillingness to admit what comes from rival camps.

On a critical examination of the definition of vyāpti, Raghunatha rejects Vyatireka vyāpti and the Kevala vyatireki type of inference. His verdict runs counter to the Nyāya standpoint. He argues that the knowledge arising from the knowledge of vyatireka concomitance is not inference at all; it is entirely different from anumiti. The instrumental cause (karaṇa) responsible for the production of this knowledge is a distinct pramāṇa, viz. Arthāpatti.[63]

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43. Ślokavārtika, 1.1.35. Tr. by Ganganāth Jhā, 87.

44. Pūrva Mīmāṁsā, 1.3.2.

45. Mimāṁsā Sūtras of Jaimini Translated by Mohan Lal Sandal, Motilal Banarsidas, Delhi, Vol.I, 1980, Pp. 23-24.

46. Ślokavārtika, I. 32-33.

47. Śaṃkara's Bhāṣya on Brahmasūtra, I.2.25.

48. Yastu na rātrau bhuṅkte nāsau divābhunjaṅatve sati pino, etc., Tarkabhāṣya (Poona Edn.) P.15.

49. Evamarthāpattirapi na pramāṇāntaram---Yadākhalvavyapakaḥ sanneketra nāsti tadanyatrāsti, etc., Tattvakaumudī, P.46.

50. Nyāyabhāṣya, I. 1.3. NB II. 2.2.

51. Nyāya Kusumāñjali, III. 19.

52. Muktāvaḷī on Kārikā, 143.

53. Tattva Cintāmaṇi.

54. Nyāya mañjarī.

55. Ibid.

56. Tattva cintamaṇi.

57. Muktāvaḷi.

58. Advaita Vedāntaparibhāṣa.

59. Nyāya Vārtika, I. 1.35.

60. Ibid.

61. Ibid.

62. Dīdhiti, on Kevalanvayi Section, Anumāna Cintāmani.

63. Dīdhiti, on Kevalānvayi section.